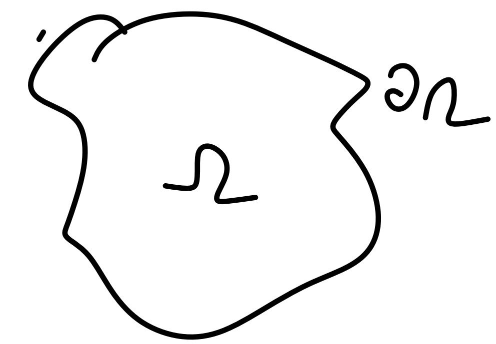
Interior Neumann problem



$$\Delta u = 0 \quad \text{in } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{on } \partial\Omega$$

Q: Is there always a solution?

A: No, solution only exists as long as

$$\int_{\partial\Omega} f = \int_{\partial\Omega} \frac{\partial u}{\partial n} = \int_{\Omega} \Delta u = 0$$

Moreover, solution only determined upto a constant

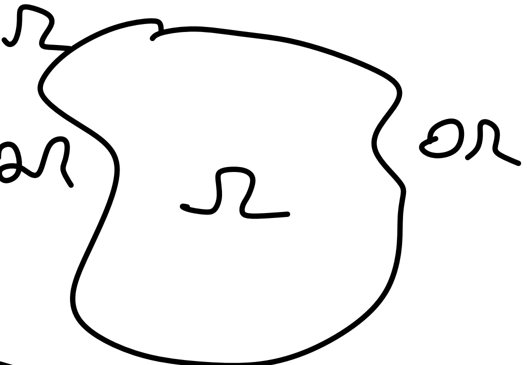
Fix: choose some normalization

Integral representation:

$u = S[\sigma](x)$

$\Delta u = 0$ in Ω

$\frac{\partial u}{\partial n} = f$ on $\partial\Omega$



$$\lim_{h \to 0^+} \nabla u(x_0 - h\, n(x_0)) \cdot n(x_0)$$

$$= \frac{1}{2}\sigma(x_0) - \overbrace{\text{p.v.}\, \frac{1}{2\pi} \int \frac{(x_0 - y)\cdot n(x_0)}{|x_0 - y|^2}\sigma(y)\, ds_y}^{S'[\sigma](x_0)} = f(x_0)$$

$$\left(\frac{I}{2} + S'\right)\sigma = f \quad \text{on } \partial\Omega$$

Note different sign of identity

HW: Show that

$$\int_{\partial\Omega} u(x)\, S'[\sigma](x)\, ds_x = \int \sigma(x)\, D^{PV}[u](x)\, ds_x$$

ie. S' & D are adjoints of one another

Integral eq^n is second kind. so are we done?

unfortunately no! If σ is a constant on $\partial\Omega$

then $\frac{1}{2}\sigma + D^{PV}\sigma = 0$ on $\partial\Omega$ Recall $N(A) = N(A^*)$

So $\left(\frac{I}{2} + S'\right)$ also has a null vector

$$\left(\frac{I}{2} + S'\right)\sigma = f$$

$\forall \sigma$ $\quad \int 1 \left(\frac{I}{2} + S'\right)\sigma = \int \sigma \left(\frac{I}{2} + D^{PV}\right)[1] = 0$

But $\int f = 0$ required for existence of sol^{n}s to PDE

So as long as data is consistent we are fine

How do we fix the null space? Add something thats not in the range!

$*$ $\quad \left(\frac{I}{2} + S'\right)\sigma + \frac{1}{|\partial\Omega|}\int_{\partial\Omega} \sigma \, ds_x = f$ $\quad$ is an invertible SKIE for σ

$$\int * = \cancel{\int \left(\frac{I}{2} + S'\right)\sigma}^{\,0} + \int \frac{1}{|\partial\Omega|}\int_{\partial\Omega} \sigma = \cancel{\int f}^{\,0}$$

$$\Rightarrow \int_{\partial\Omega} \sigma = 0$$

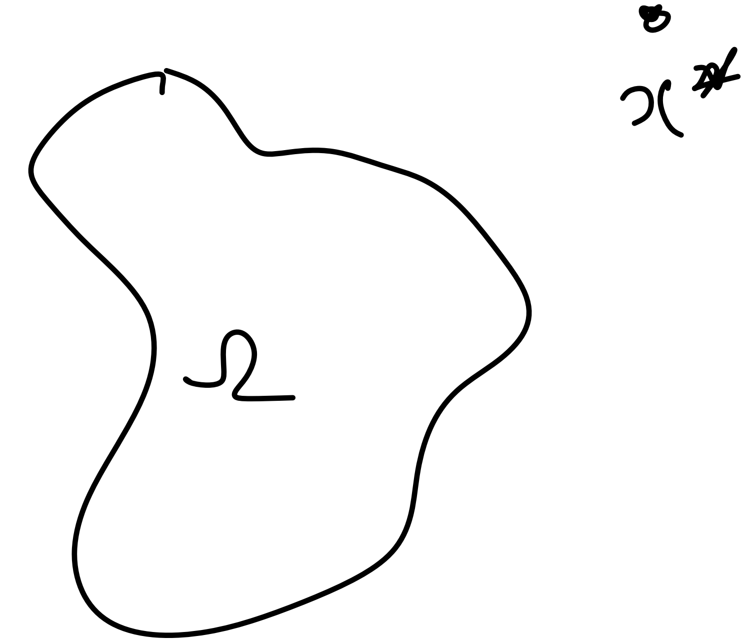
So if σ solves $*$ it also solves
$\left(\frac{I}{2} + S'\right)\sigma = f$ & $\int \sigma = 0$

Details on building a numerical test:

Consider INP again

$$\Delta u = 0 \quad \text{in } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{in } \partial\Omega$$



Step 1: Construct a known sol^n to Laplace's equation in Ω

say $u_{ex}(x) = G(x, x^*)$

Suppose now that $f(x) = \frac{\partial u^{ex}}{\partial n}(x, x^*) = \frac{\partial G}{\partial n}(x, x^*)$ for $x \in \partial\Omega$

then any solution to INP with this data must satisfy

$$u(x) = u_{ex}(x) + c \quad \text{for a fixed constant } c$$

Analytic solution test :

Consider INP again

$\Delta u = 0$ in Ω

$\frac{\partial u}{\partial n} = f$ in $\partial\Omega$

$f = \frac{\partial G}{\partial n}(x, x^*)$ on $\partial\Omega$

$u_{ex} = G(x, x^*)$ in Ω

Step2: Solve INP using our integral equation, i.e.

$u(x) = S[\sigma](x)$ &

σ solves

$$\frac{1}{2}\sigma(x) + S'[\sigma](x) + \frac{1}{|\partial\Omega|}\int_{\partial\Omega}\sigma = f(x) \text{ on } \partial\Omega$$

Step3: After computing σ by solving IE above

Evaluate $u(x_j) = S[\sigma](x_j) = \int_{\partial\Omega} G(x_j, y)\sigma(y)\,ds_y \quad j = 1, 2$

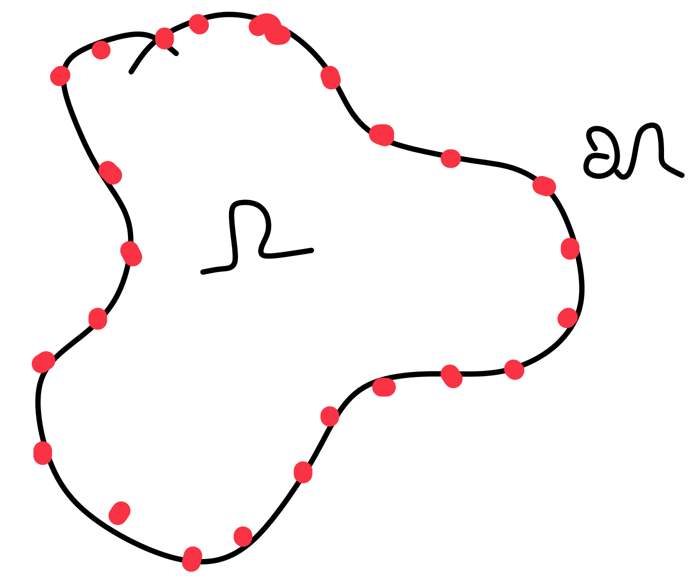
Step4 : Verify

$u(x_1) - u(x_2) = u_{ex}(x_1) + c - (u_{ex}(x_2) + c) = u_{ex}(x_1) - u_{ex}(x_2)$

Integral equation → linear system

$$\sigma(x) + \int_{\partial\Omega} K(x,y)\,\sigma(y)\,ds_y = f(y)$$

K smooth in both x, y.



Suppose $\gamma(t) : [-\pi, \pi) \to \partial\Omega$

$\gamma(t)$ periodic since curve is closed.

Let $x_j \in \mathbb{R}^2 = \gamma(t_j)$ $\quad t_j = \frac{2\pi j}{2n+1}$

& $\sigma_j = \sigma(x_j)$ $\quad j = -n, \ldots, n$

$f_j = f(x_j)$

← Discrete representation for σ, f & $\partial\Omega$

Also need ability to compute

$$\int K(x,y)\,\sigma(y)$$

Quadrature rules

$$\int_{-\pi}^{\pi} f(t) \approx \sum f(t_j) w_j$$

(t_j, w_j)

nodes

weights

is called a quadrature rule

Midpoint rule / periodic trapezoid rule

$$\int_{-\pi}^{\pi} f(t) \approx \frac{2\pi}{2n+1} \sum_{j=-n}^{n} f(t_j)$$

$$t_j = \frac{2\pi j}{2n+1} \qquad w_j = \frac{2\pi}{2n+1}$$

Q: what is the error?

$$f(t) = \sum_{m=-\infty}^{\infty} \hat{f}_m e^{imt}$$

$$\int_{-\pi}^{\pi} e^{imt} dt = 0 \quad \text{if } m \neq 0$$

$$= 2\pi \quad \text{if } m = 0$$

$$\frac{2\pi}{2n+1} \sum_{j=-n}^{n} e^{imt_j} =$$

$$\frac{2\pi}{2n+1} \sum_{j=-n}^{n} e^{im\frac{2\pi j}{2n+1}}$$

$= 1$ if $m \bmod 2n+1 = 0$

$= 0$ otherwise

Error analysis periodic trapezoid rule

$$\int_{-\pi}^{\pi} f(t)\,dt = \int_{-\pi}^{\pi} \sum_{m=-\infty}^{\infty} \hat{f}_m e^{imt}\,dt = \hat{f}_0$$

$$\frac{2\pi}{2n+1} \sum_{j=-n}^{n} f(t_j) = \frac{2\pi}{2n+1} \sum_{j=-n}^{n} \sum_{m=-\infty}^{\infty} \hat{f}_m e^{imt_j}$$

$$= \hat{f}_0 + \sum_{\substack{m=-\infty \\ m\neq 0}}^{\infty} \hat{f}_{(2n+1)m}$$

$$\therefore \left| \int_{-\pi}^{\pi} f(t)\,dt - \frac{2\pi}{2n+1} \sum_{j=-n}^{n} f(t_j) \right| = \left| \sum_{\substack{m=-\infty \\ m\neq 0}}^{\infty} \hat{f}_{(2n+1)m} \right|$$

$= O(n^{-(k-1)})$ for C^k functions

$= O(e^{-an})$ for analytic functions

Much better than standard error estimate $O(h^2)$

Need periodicity + smoothness for better rate

$$\sigma(x) + \int_{\partial\Omega} K(x,y)\sigma(y)\, ds_y = f(x) \qquad x \in \partial\Omega$$

$$\sigma(x) + \int_{-\pi}^{\pi} K(x, \gamma(t))\, \sigma(\gamma(t))\, |\gamma'(t)|\, dt = f(x) \qquad x \in \partial\Omega$$

Nyström method: Replace integral by quadrature rule & enforce equation at quadrature nodes

$$\sigma(x) + \sum_{j=-n}^{n} K(x, \gamma(t_j))\, |\gamma'(t_j)|\, w_j \sigma_j = f(x) \qquad x \in \partial\Omega$$

$x = \gamma(t_m) = x_m$

$$\sigma_m + \sum_{j=-n}^{n} K(\gamma(t_m), \gamma(t_j))\, |\gamma'(t_j)|\, w_j \sigma_j = f_m$$

$$A_{m,j} = \delta_{m,j} + K(\gamma(t_m), \gamma(t_j))\, |\gamma'(t_j)|\, w_j$$

Nice property of SKIE   After computing σ_m, σ on rest of domain given by

$$\sigma(x) = f(x) - \int K(x,y)\,\sigma(y)\,d_y$$

$$\sigma(x) = f(x) - \sum_{j=-n}^{n} K(x, \gamma(t_j))\,\sigma_j\,|\gamma'(t_j)|\,w_j$$

Laplace Kernels

D: Double layer potential $\dfrac{(x-y)\cdot n(y)}{2\pi\,|x-y|^2} = K_D$

S': Normal derivative of single layer $-\dfrac{(x-y)\cdot n(x)}{2\pi\,|x-y|^2} = K_{S'}$

$$\lim_{x \to y} K_D(x,y) = \frac{\kappa(x)}{2\pi}$$

PTR demo

Other kernels

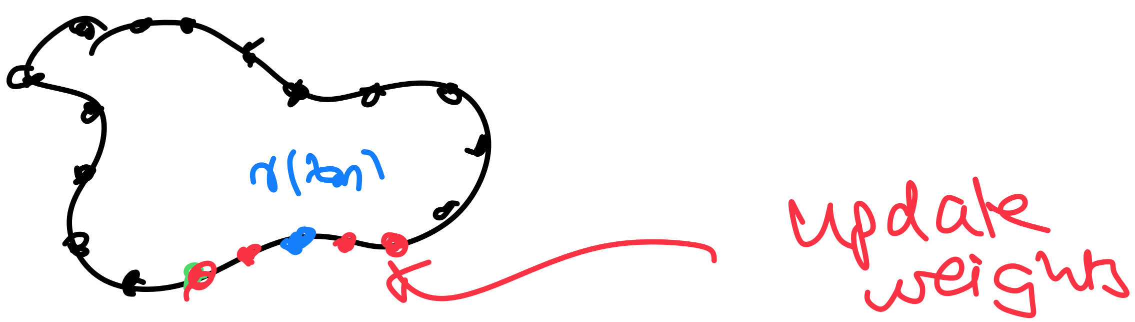
Helmholtz double layer potential $(\Delta + k^2)u = 0$

$$K_{D_n} = \frac{i}{4}\frac{\partial}{\partial n} H_0^{(1)}(k|x-y|)$$

$$K_{D_n}(x,y) = O(\log|x-y|)$$

as $x \to y$

singular on diagonal



Many options available

Product quadrature rules (Kress, Helsing)

Quadrature by expansion

singularity subtraction

Locally corrected GGQ based rules

$$\sum_{j=-n}^{n} K(\gamma(t_m), \gamma(t_j))\, \sigma_j w_j$$

$$\to \sum_{m,j \text{ close}} K(\gamma(t_m), \gamma(t_j))\, \sigma_j w_{jm} + \sum_{m,j \text{ far}} K(\gamma(t_m), \gamma(t_j))\, \sigma_j w_j$$

(Fast algorithm compatible!)